

Talha Mahamud

Research Assistant, Spintronics Lab · Physics Discipline, Khulna University

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RESEARCH INTERESTS

- Altermagnetism
- Spin-Orbit Torque (SOT) Spintronics
- Skyrmions
- Magnonics
- Topological Spintronics
- Physics-Driven Machine Learning
- Quantum Computation

EDUCATION

Bachelor of Science in Physics

Khulna University, Khulna · CGPA: 3.34 / 4.00

Feb 2022 – Dec 2025

Higher Secondary School Certificate

Al Hajj Sarwar Khan College, Khulna · GPA: 5.00 / 5.00

2018 – 2020

RESEARCH EXPERIENCE

Research Assistant

Spintronics Lab, Physics Discipline, Khulna University

Jan 2026 – Present

PUBLICATION

T. Mahamud, S. Rahman, A. A. Nahid, “Ensemble Learning Based Pulsar Detection Along with Feature Engineering,” *Int. J. Advanced Networking and Applications*, vol. 17, no. 3, pp. 6916–6921, 2025. ISSN: 0975-0290. [\[PDF\]](#)

TECHNICAL SKILLS

Micromagnetic Simulation	Mumax3, VAMPIRE
Programming	Python, C, C++, MATLAB, $\text{\LaTeX}$
Web Development	HTML, CSS, Flask
Operating Systems	Linux, Windows
Office & Graphics	MS Word, PowerPoint, Excel; Adobe Illustrator, Blender

ADDITIONAL COURSES

Scientific Computing with Python

freeCodeCamp

Jan 2023 – Nov 2023

Supervised Machine Learning: Regression and Classification

Coursera · Instructor: Andrew Ng

Jun 2023 – Oct 2023

Advanced Learning Algorithms

Coursera · Instructor: Andrew Ng

May 2024 – Jul 2024

Differential Equations

Instructor: Prof. Daniel Gabilondo, Seminole State College, Florida

Jan 2024 – Mar 2024

Linear Algebra

Instructor: Prof. Gilbert Strang, Massachusetts Institute of Technology

Feb 2023 – Jul 2023

AWARDS & ACHIEVEMENTS

University Merit Scholarship Physics Discipline, Khulna University	2023, 2024
NASA Space Apps Challenge Shortlisted among the top 50 projects (BASIS)	Aug 2024
University Physics Competition Competitor	Nov 2022

TEACHING & LEADERSHIP

Fermion Physics Club, Khulna University Instructor and Club Administrator	2022 – 2024
Workshop on MATLAB & Python Libraries Organised by Fermion Physics Club, Khulna University	Sep 2024
Python for Beginners Organised by Fermion Physics Club, Khulna University	May 2023

SCIENCE WRITING

Content Writer, Substack & Medium Natural science, machine learning, philosophy, biography	2025 – Present
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WORKSHOPS & SEMINARS ATTENDED

3rd Workshop on Relativity Organised by Community of Physics	Oct 2022
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LANGUAGES

Bengali	Native
English	Proficient
German	Beginner

REFEREES

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